

Class 09; Candy Mini Project

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Background

Today we are taking a wee detour to analyze a fun data-set (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

Data Import

```
candy <- read.csv("candy-data.txt", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0

Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

There are 85 different candy types.

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

For fruity candy types, there are 38 in the dataset.

```
sum(candy$fruity)
```

```
sum(candy$fruity)
```

```
[1] 38
```

What is your favorite Candy?

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My other favorite candy is Haribo Sour Bears, these winpercent value is 51.41243.

```
candy["Haribo Sour Bears", ]$winpercent
```

```
[1] 51.41243
```

Q4. What is the winpercent value for "Kit Kat"?

The winpercent value for Kit Kat is 76.7686.

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

The win percent value for Tootsie Rolls is 49.6535.

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```
library(skimr)
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, columns for sugarpercent, pricepercent, and winpercent are on a different scale because they contain continuous decimal values where others are just values of 0 and 1.

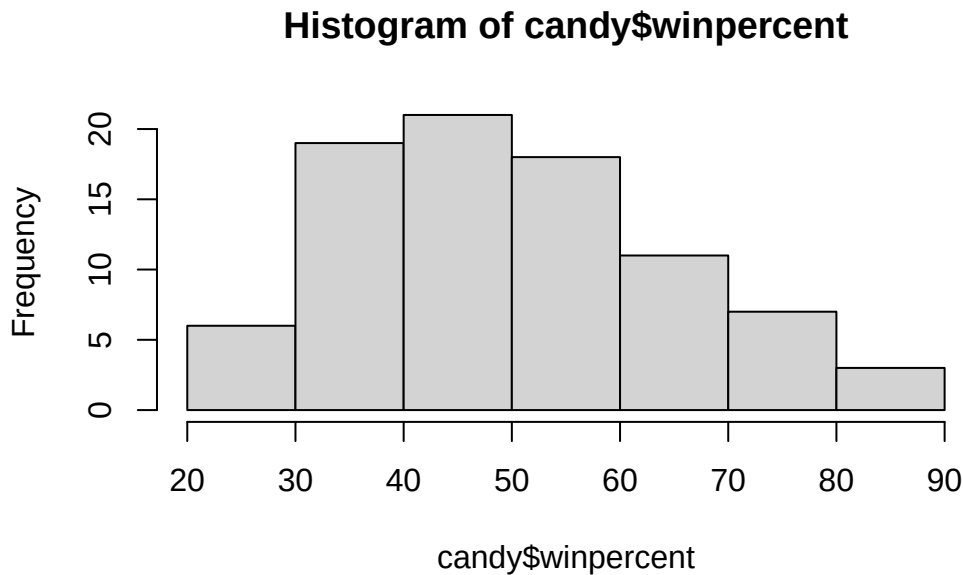
Q7. What do you think a zero and one represent for the candy\$chocolate column?

In the candy\$chocolate column, 0 means that the candy doesn't contain chocolate and 1 means yes it does contain.

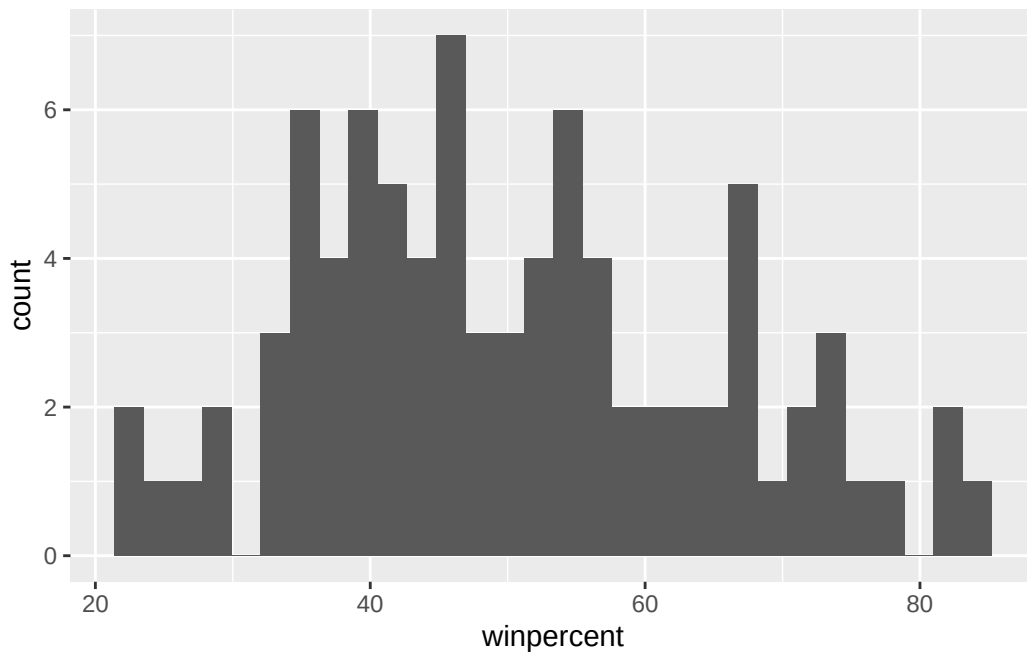
##Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent)
```



```
library(ggplot2)
ggplot(candy) + aes(winpercent) + geom_histogram(bins = 30)
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values seem slightly skewed to the right.

Q10. Is the center of the distribution above or below 50%?

It is slightly higher than 50%, the mean winpercent is approximately 50.31%.

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Chocolate candy is ranked higher on average since it's mean winpercent (60.921) is greater than fruit candy (44.12).

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy [choc.ind, ]

choc.win <- choc.candy$winpercent
choc.win
```

```
[1] 66.97173 67.60294 50.34755 56.91455 38.97504 55.37545 62.28448 56.49050
[9] 59.23612 57.21925 76.76860 71.46505 66.57458 55.06407 73.09956 60.80070
[17] 64.35334 47.82975 54.52645 70.73564 66.47068 69.48379 81.86626 84.18029
[25] 73.43499 72.88790 65.71629 34.72200 37.88719 76.67378 59.52925 48.98265
[33] 43.06890 45.73675 49.65350 81.64291 49.52411
```

```
mean (choc.win)
```

```
[1] 60.92153
```

```
fruit.ind <- as.logical (candy$fruity)
fruit.candy <- candy [fruit.ind,]
fruit.win <- fruit.candy$winpercent
mean (fruit.win)
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

Yes the difference is statistically significant because the p-value (4.164×10^{-10}) is less than 0.05.

```
t.test(winpercent ~ chocolate, data = candy)
```

Welch Two Sample t-test

```
data: winpercent by chocolate
t = -7.3031, df = 67.539, p-value = 4.164e-10
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -23.91110 -13.64744
sample estimates:
mean in group 0 mean in group 1
    42.14226      60.92153
```

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

The five least liked candy types include Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, Jawbusters.

```
candy[order(candy$winpercent), ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0
Root Beer Barrels	0	0	0	0	0
Sugar Daddy	0	0	1	0	0
One dime	0	0	0	0	0
Sugar Babies	0	0	1	0	0
Haribo Happy Cola	0	0	0	0	0
Caramel Apple Pops	0	1	1	0	0
Strawberry bon bons	0	1	0	0	0
Sixlets	1	0	0	0	0
Ring pop	0	1	0	0	0
Chewey Lemonhead Fruit Mix	0	1	0	0	0
Red vines	0	1	0	0	0
Pixie Sticks	0	0	0	0	0
Nestle Smarties	1	0	0	0	0
Candy Corn	0	0	0	0	0
Charleston Chew	1	0	0	0	1
Warheads	0	1	0	0	0
Lemonhead	0	1	0	0	0
Fun Dip	0	1	0	0	0
Now & Later	0	1	0	0	0
Dum Dums	0	1	0	0	0
Pop Rocks	0	1	0	0	0
Laffy Taffy	0	1	0	0	0
Werther's Original Caramel	0	0	1	0	0
Haribo Twin Snakes	0	1	0	0	0
Dots	0	1	0	0	0
Runts	0	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0

Fruit Chews	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
Twizzlers	0	1	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Smarties candy	0	1	0	0	0
One quarter	0	0	0	0	0
Payday	0	0	0	1	1
Mike & Ike	0	1	0	0	0
Gobstopper	0	1	0	0	0
Trolli Sour Bites	0	1	0	0	0
Mounds	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Whoppers	1	0	0	0	0
Tootsie Roll Snack Bars	1	0	0	0	0
Almond Joy	1	0	0	1	0
Haribo Sour Bears	0	1	0	0	0
Air Heads	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Lifesavers big ring gummies	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Swedish Fish	0	1	0	0	0
Milk Duds	1	0	1	0	0
Skittles wildberry	0	1	0	0	0
Nerds	0	1	0	0	0
Hershey's Kisses	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
Baby Ruth	1	0	1	1	1
Haribo Gold Bears	0	1	0	0	0
Junior Mints	1	0	0	0	0
Hershey's Special Dark	1	0	0	0	0
Snickers Crisper	1	0	1	1	0
Sour Patch Kids	0	1	0	0	0
Milky Way Midnight	1	0	1	0	1
Hershey's Krackel	1	0	0	0	0
Skittles original	0	1	0	0	0
Milky Way Simply Caramel	1	0	1	0	0
Rolo	1	0	1	0	0
Nestle Crunch	1	0	0	0	0
M&M's	1	0	0	0	0
100 Grand	1	0	1	0	0
Starburst	0	1	0	0	0
3 Musketeers	1	0	0	0	1
Peanut M&Ms	1	0	0	1	0

Nestle Butterfinger	1	0	0	1	0	
Peanut butter M&M's	1	0	0	1	0	
Reese's stuffed with pieces	1	0	0	1	0	
Milky Way	1	0	1	0	1	
Reese's pieces	1	0	0	1	0	
Snickers	1	0	1	1	1	
Kit Kat	1	0	0	0	0	
Twix	1	0	1	0	0	
Reese's Miniatures	1	0	0	1	0	
Reese's Peanut Butter cup	1	0	0	1	0	
	crisped	ricewafer	hard bar	pluribus	sugarpercent	
Nik L Nip		0	0	0	1	0.197
Boston Baked Beans		0	0	0	1	0.313
Chiclets		0	0	0	1	0.046
Super Bubble		0	0	0	0	0.162
Jawbusters		0	1	0	1	0.093
Root Beer Barrels		0	1	0	1	0.732
Sugar Daddy		0	0	0	0	0.418
One dime		0	0	0	0	0.011
Sugar Babies		0	0	0	1	0.965
Haribo Happy Cola		0	0	0	1	0.465
Caramel Apple Pops		0	0	0	0	0.604
Strawberry bon bons		0	1	0	1	0.569
Sixlets		0	0	0	1	0.220
Ring pop		0	1	0	0	0.732
Chewey Lemonhead Fruit Mix		0	0	0	1	0.732
Red vines		0	0	0	1	0.581
Pixie Sticks		0	0	0	1	0.093
Nestle Smarties		0	0	0	1	0.267
Candy Corn		0	0	0	1	0.906
Charleston Chew		0	0	1	0	0.604
Warheads		0	1	0	0	0.093
Lemonhead		0	1	0	0	0.046
Fun Dip		0	1	0	0	0.732
Now & Later		0	0	0	1	0.220
Dum Dums		0	1	0	0	0.732
Pop Rocks		0	1	0	1	0.604
Laffy Taffy		0	0	0	0	0.220
Werther's Original Caramel		0	1	0	0	0.186
Haribo Twin Snakes		0	0	0	1	0.465
Dots		0	0	0	1	0.732
Runts		0	1	0	1	0.872
Tootsie Roll Juniors		0	0	0	0	0.313

Fruit Chews	0	0	0	1	0.127
Welch's Fruit Snacks	0	0	0	1	0.313
Twizzlers	0	0	0	0	0.220
Tootsie Roll Midgies	0	0	0	1	0.174
Smarties candy	0	1	0	1	0.267
One quarter	0	0	0	0	0.011
Payday	0	0	1	0	0.465
Mike & Ike	0	0	0	1	0.872
Gobstopper	0	1	0	1	0.906
Trolli Sour Bites	0	0	0	1	0.313
Mounds	0	0	1	0	0.313
Tootsie Pop	0	1	0	0	0.604
Whoppers	1	0	0	1	0.872
Tootsie Roll Snack Bars	0	0	1	0	0.465
Almond Joy	0	0	1	0	0.465
Haribo Sour Bears	0	0	0	1	0.465
Air Heads	0	0	0	0	0.906
Sour Patch Tricksters	0	0	0	1	0.069
Lifesavers big ring gummies	0	0	0	0	0.267
Mr Good Bar	0	0	1	0	0.313
Swedish Fish	0	0	0	1	0.604
Milk Duds	0	0	0	1	0.302
Skittles wildberry	0	0	0	1	0.941
Nerds	0	1	0	1	0.848
Hershey's Kisses	0	0	0	1	0.127
Hershey's Milk Chocolate	0	0	1	0	0.430
Baby Ruth	0	0	1	0	0.604
Haribo Gold Bears	0	0	0	1	0.465
Junior Mints	0	0	0	1	0.197
Hershey's Special Dark	0	0	1	0	0.430
Snickers Crisper	1	0	1	0	0.604
Sour Patch Kids	0	0	0	1	0.069
Milky Way Midnight	0	0	1	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Skittles original	0	0	0	1	0.941
Milky Way Simply Caramel	0	0	1	0	0.965
Rolo	0	0	0	1	0.860
Nestle Crunch	1	0	1	0	0.313
M&M's	0	0	0	1	0.825
100 Grand	1	0	1	0	0.732
Starburst	0	0	0	1	0.151
3 Musketeers	0	0	1	0	0.604
Peanut M&Ms	0	0	0	1	0.593

Nestle Butterfinger	0	0	1	0	0.604
Peanut butter M&M's	0	0	0	1	0.825
Reese's stuffed with pieces	0	0	0	0	0.988
Milky Way	0	0	1	0	0.604
Reese's pieces	0	0	0	1	0.406
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Boston Baked Beans	0.511	23.41782
Chiclets	0.325	24.52499
Super Bubble	0.116	27.30386
Jawbusters	0.511	28.12744
Root Beer Barrels	0.069	29.70369
Sugar Daddy	0.325	32.23100
One dime	0.116	32.26109
Sugar Babies	0.767	33.43755
Haribo Happy Cola	0.465	34.15896
Caramel Apple Pops	0.325	34.51768
Strawberry bon bons	0.058	34.57899
Sixlets	0.081	34.72200
Ring pop	0.965	35.29076
Chewey Lemonhead Fruit Mix	0.511	36.01763
Red vines	0.116	37.34852
Pixie Sticks	0.023	37.72234
Nestle Smarties	0.976	37.88719
Candy Corn	0.325	38.01096
Charleston Chew	0.511	38.97504
Warheads	0.116	39.01190
Lemonhead	0.104	39.14106
Fun Dip	0.325	39.18550
Now & Later	0.325	39.44680
Dum Dums	0.034	39.46056
Pop Rocks	0.837	41.26551
Laffy Taffy	0.116	41.38956
Werther's Original Caramel	0.267	41.90431
Haribo Twin Snakes	0.465	42.17877
Dots	0.511	42.27208
Runts	0.279	42.84914
Tootsie Roll Juniors	0.511	43.06890

Fruit Chews	0.034	43.08892
Welch's Fruit Snacks	0.313	44.37552
Twizzlers	0.116	45.46628
Tootsie Roll Midgies	0.011	45.73675
Smarties candy	0.116	45.99583
One quarter	0.511	46.11650
Payday	0.767	46.29660
Mike & Ike	0.325	46.41172
Gobstopper	0.453	46.78335
Trolli Sour Bites	0.255	47.17323
Mounds	0.860	47.82975
Tootsie Pop	0.325	48.98265
Whoppers	0.848	49.52411
Tootsie Roll Snack Bars	0.325	49.65350
Almond Joy	0.767	50.34755
Haribo Sour Bears	0.465	51.41243
Air Heads	0.511	52.34146
Sour Patch Tricksters	0.116	52.82595
Lifesavers big ring gummies	0.279	52.91139
Mr Good Bar	0.918	54.52645
Swedish Fish	0.755	54.86111
Milk Duds	0.511	55.06407
Skittles wildberry	0.220	55.10370
Nerds	0.325	55.35405
Hershey's Kisses	0.093	55.37545
Hershey's Milk Chocolate	0.918	56.49050
Baby Ruth	0.767	56.91455
Haribo Gold Bears	0.465	57.11974
Junior Mints	0.511	57.21925
Hershey's Special Dark	0.918	59.23612
Snickers Crisper	0.651	59.52925
Sour Patch Kids	0.116	59.86400
Milky Way Midnight	0.441	60.80070
Hershey's Krackel	0.918	62.28448
Skittles original	0.220	63.08514
Milky Way Simply Caramel	0.860	64.35334
Rolo	0.860	65.71629
Nestle Crunch	0.767	66.47068
M&M's	0.651	66.57458
100 Grand	0.860	66.97173
Starburst	0.220	67.03763
3 Musketeers	0.511	67.60294
Peanut M&Ms	0.651	69.48379

Nestle Butterfinger	0.767	70.73564
Peanut butter M&M's	0.651	71.46505
Reese's stuffed with pieces	0.651	72.88790
Milky Way	0.651	73.09956
Reese's pieces	0.651	73.43499
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

Q14. What are the top 5 all time favorite candy types out of this set?

The top 5 all time favorites is Reese's Peanut Butter Cup, Reese's Miniatures, Twix, and Kit Kats.

```
candy[order(-candy$winpercent), ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1
Reese's pieces	1	0	0	1	0
Milky Way	1	0	1	0	1
Reese's stuffed with pieces	1	0	0	1	0
Peanut butter M&M's	1	0	0	1	0
Nestle Butterfinger	1	0	0	1	0
Peanut M&Ms	1	0	0	1	0
3 Musketeers	1	0	0	0	1
Starburst	0	1	0	0	0
100 Grand	1	0	1	0	0
M&M's	1	0	0	0	0
Nestle Crunch	1	0	0	0	0
Rolo	1	0	1	0	0
Milky Way Simply Caramel	1	0	1	0	0
Skittles original	0	1	0	0	0
Hershey's Krackel	1	0	0	0	0
Milky Way Midnight	1	0	1	0	1
Sour Patch Kids	0	1	0	0	0
Snickers Crisper	1	0	1	1	0
Hershey's Special Dark	1	0	0	0	0

Junior Mints	1	0	0	0	0
Haribo Gold Bears	0	1	0	0	0
Baby Ruth	1	0	1	1	1
Hershey's Milk Chocolate	1	0	0	0	0
Hershey's Kisses	1	0	0	0	0
Nerds	0	1	0	0	0
Skittles wildberry	0	1	0	0	0
Milk Duds	1	0	1	0	0
Swedish Fish	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Lifesavers big ring gummies	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Air Heads	0	1	0	0	0
Haribo Sour Bears	0	1	0	0	0
Almond Joy	1	0	0	1	0
Tootsie Roll Snack Bars	1	0	0	0	0
Whoppers	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Mounds	1	0	0	0	0
Trolli Sour Bites	0	1	0	0	0
Gobstopper	0	1	0	0	0
Mike & Ike	0	1	0	0	0
Payday	0	0	0	1	1
One quarter	0	0	0	0	0
Smarties candy	0	1	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Twizzlers	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
Fruit Chews	0	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0
Runts	0	1	0	0	0
Dots	0	1	0	0	0
Haribo Twin Snakes	0	1	0	0	0
Werther's Original Caramel	0	0	1	0	0
Laffy Taffy	0	1	0	0	0
Pop Rocks	0	1	0	0	0
Dum Dums	0	1	0	0	0
Now & Later	0	1	0	0	0
Fun Dip	0	1	0	0	0
Lemonhead	0	1	0	0	0
Warheads	0	1	0	0	0
Charleston Chew	1	0	0	0	1
Candy Corn	0	0	0	0	0

Nestle Smarties	1	0	0	0	0
Pixie Sticks	0	0	0	0	0
Red vines	0	1	0	0	0
Chewey Lemonhead Fruit Mix	0	1	0	0	0
Ring pop	0	1	0	0	0
Sixlets	1	0	0	0	0
Strawberry bon bons	0	1	0	0	0
Caramel Apple Pops	0	1	1	0	0
Haribo Happy Cola	0	0	0	0	0
Sugar Babies	0	0	1	0	0
One dime	0	0	0	0	0
Sugar Daddy	0	0	1	0	0
Root Beer Barrels	0	0	0	0	0
Jawbusters	0	1	0	0	0
Super Bubble	0	1	0	0	0
Chiclets	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Nik L Nip	0	1	0	0	0
	crisped	ricewafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0.720
Reese's Miniatures		0	0	0	0.034
Twix		1	0	1	0.546
Kit Kat		1	0	1	0.313
Snickers		0	0	1	0.546
Reese's pieces		0	0	0	1.0406
Milky Way		0	0	1	0.604
Reese's stuffed with pieces		0	0	0	0.988
Peanut butter M&M's		0	0	0	1.0825
Nestle Butterfinger		0	0	1	0.604
Peanut M&Ms		0	0	0	1.0593
3 Musketeers		0	0	1	0.604
Starburst		0	0	0	1.0151
100 Grand		1	0	1	0.732
M&M's		0	0	0	1.0825
Nestle Crunch		1	0	1	0.313
Rolo		0	0	0	1.0860
Milky Way Simply Caramel		0	0	1	0.965
Skittles original		0	0	0	1.0941
Hershey's Krackel		1	0	1	0.430
Milky Way Midnight		0	0	1	0.732
Sour Patch Kids		0	0	0	1.0069
Snickers Crisper		1	0	1	0.604
Hershey's Special Dark		0	0	1	0.430

Junior Mints	0	0	0	1	0.197
Haribo Gold Bears	0	0	0	1	0.465
Baby Ruth	0	0	1	0	0.604
Hershey's Milk Chocolate	0	0	1	0	0.430
Hershey's Kisses	0	0	0	1	0.127
Nerds	0	1	0	1	0.848
Skittles wildberry	0	0	0	1	0.941
Milk Duds	0	0	0	1	0.302
Swedish Fish	0	0	0	1	0.604
Mr Good Bar	0	0	1	0	0.313
Lifesavers big ring gummies	0	0	0	0	0.267
Sour Patch Tricksters	0	0	0	1	0.069
Air Heads	0	0	0	0	0.906
Haribo Sour Bears	0	0	0	1	0.465
Almond Joy	0	0	1	0	0.465
Tootsie Roll Snack Bars	0	0	1	0	0.465
Whoppers	1	0	0	1	0.872
Tootsie Pop	0	1	0	0	0.604
Mounds	0	0	1	0	0.313
Trolli Sour Bites	0	0	0	1	0.313
Gobstopper	0	1	0	1	0.906
Mike & Ike	0	0	0	1	0.872
Payday	0	0	1	0	0.465
One quarter	0	0	0	0	0.011
Smarties candy	0	1	0	1	0.267
Tootsie Roll Midgies	0	0	0	1	0.174
Twizzlers	0	0	0	0	0.220
Welch's Fruit Snacks	0	0	0	1	0.313
Fruit Chews	0	0	0	1	0.127
Tootsie Roll Juniors	0	0	0	0	0.313
Runts	0	1	0	1	0.872
Dots	0	0	0	1	0.732
Haribo Twin Snakes	0	0	0	1	0.465
Werther's Original Caramel	0	1	0	0	0.186
Laffy Taffy	0	0	0	0	0.220
Pop Rocks	0	1	0	1	0.604
Dum Dums	0	1	0	0	0.732
Now & Later	0	0	0	1	0.220
Fun Dip	0	1	0	0	0.732
Lemonhead	0	1	0	0	0.046
Warheads	0	1	0	0	0.093
Charleston Chew	0	0	1	0	0.604
Candy Corn	0	0	0	1	0.906

Nestle Smarties	0	0	0	1	0.267
Pixie Sticks	0	0	0	1	0.093
Red vines	0	0	0	1	0.581
Chewey Lemonhead Fruit Mix	0	0	0	1	0.732
Ring pop	0	1	0	0	0.732
Sixlets	0	0	0	1	0.220
Strawberry bon bons	0	1	0	1	0.569
Caramel Apple Pops	0	0	0	0	0.604
Haribo Happy Cola	0	0	0	1	0.465
Sugar Babies	0	0	0	1	0.965
One dime	0	0	0	0	0.011
Sugar Daddy	0	0	0	0	0.418
Root Beer Barrels	0	1	0	1	0.732
Jawbusters	0	1	0	1	0.093
Super Bubble	0	0	0	0	0.162
Chiclets	0	0	0	1	0.046
Boston Baked Beans	0	0	0	1	0.313
Nik L Nip	0	0	0	1	0.197

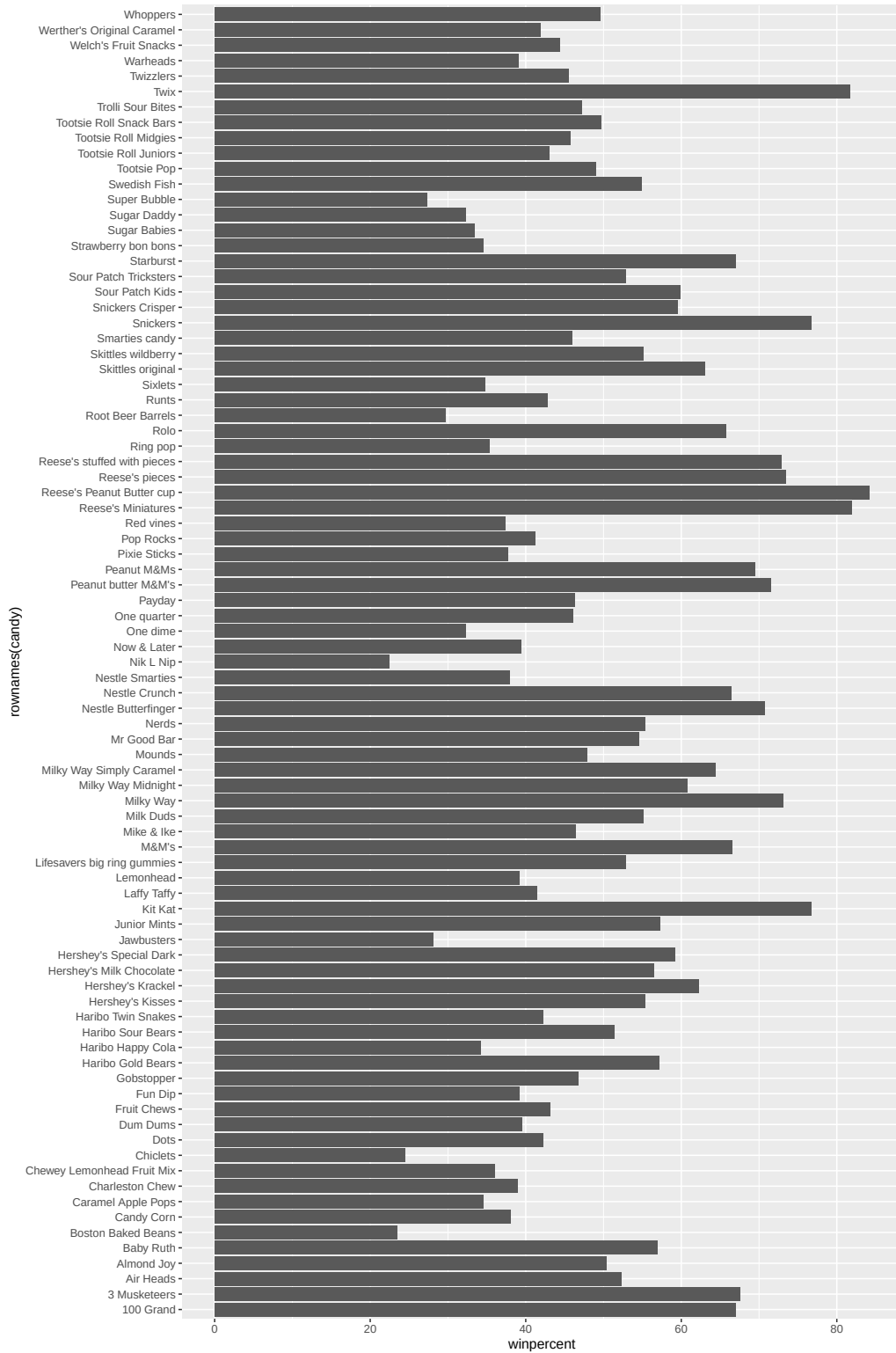
	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378
Reese's pieces	0.651	73.43499
Milky Way	0.651	73.09956
Reese's stuffed with pieces	0.651	72.88790
Peanut butter M&M's	0.651	71.46505
Nestle Butterfinger	0.767	70.73564
Peanut M&Ms	0.651	69.48379
3 Musketeers	0.511	67.60294
Starburst	0.220	67.03763
100 Grand	0.860	66.97173
M&M's	0.651	66.57458
Nestle Crunch	0.767	66.47068
Rolo	0.860	65.71629
Milky Way Simply Caramel	0.860	64.35334
Skittles original	0.220	63.08514
Hershey's Krackel	0.918	62.28448
Milky Way Midnight	0.441	60.80070
Sour Patch Kids	0.116	59.86400
Snickers Crisper	0.651	59.52925
Hershey's Special Dark	0.918	59.23612

Junior Mints	0.511	57.21925
Haribo Gold Bears	0.465	57.11974
Baby Ruth	0.767	56.91455
Hershey's Milk Chocolate	0.918	56.49050
Hershey's Kisses	0.093	55.37545
Nerds	0.325	55.35405
Skittles wildberry	0.220	55.10370
Milk Duds	0.511	55.06407
Swedish Fish	0.755	54.86111
Mr Good Bar	0.918	54.52645
Lifesavers big ring gummies	0.279	52.91139
Sour Patch Tricksters	0.116	52.82595
Air Heads	0.511	52.34146
Haribo Sour Bears	0.465	51.41243
Almond Joy	0.767	50.34755
Tootsie Roll Snack Bars	0.325	49.65350
Whoppers	0.848	49.52411
Tootsie Pop	0.325	48.98265
Mounds	0.860	47.82975
Trolli Sour Bites	0.255	47.17323
Gobstopper	0.453	46.78335
Mike & Ike	0.325	46.41172
Payday	0.767	46.29660
One quarter	0.511	46.11650
Smarties candy	0.116	45.99583
Tootsie Roll Midgies	0.011	45.73675
Twizzlers	0.116	45.46628
Welch's Fruit Snacks	0.313	44.37552
Fruit Chews	0.034	43.08892
Tootsie Roll Juniors	0.511	43.06890
Runts	0.279	42.84914
Dots	0.511	42.27208
Haribo Twin Snakes	0.465	42.17877
Werther's Original Caramel	0.267	41.90431
Laffy Taffy	0.116	41.38956
Pop Rocks	0.837	41.26551
Dum Dums	0.034	39.46056
Now & Later	0.325	39.44680
Fun Dip	0.325	39.18550
Lemonhead	0.104	39.14106
Warheads	0.116	39.01190
Charleston Chew	0.511	38.97504
Candy Corn	0.325	38.01096

Nestle Smarties	0.976	37.88719
Pixie Sticks	0.023	37.72234
Red vines	0.116	37.34852
Chewey Lemonhead Fruit Mix	0.511	36.01763
Ring pop	0.965	35.29076
Sixlets	0.081	34.72200
Strawberry bon bons	0.058	34.57899
Caramel Apple Pops	0.325	34.51768
Haribo Happy Cola	0.465	34.15896
Sugar Babies	0.767	33.43755
One dime	0.116	32.26109
Sugar Daddy	0.325	32.23100
Root Beer Barrels	0.069	29.70369
Jawbusters	0.511	28.12744
Super Bubble	0.116	27.30386
Chiclets	0.325	24.52499
Boston Baked Beans	0.511	23.41782
Nik L Nip	0.976	22.44534

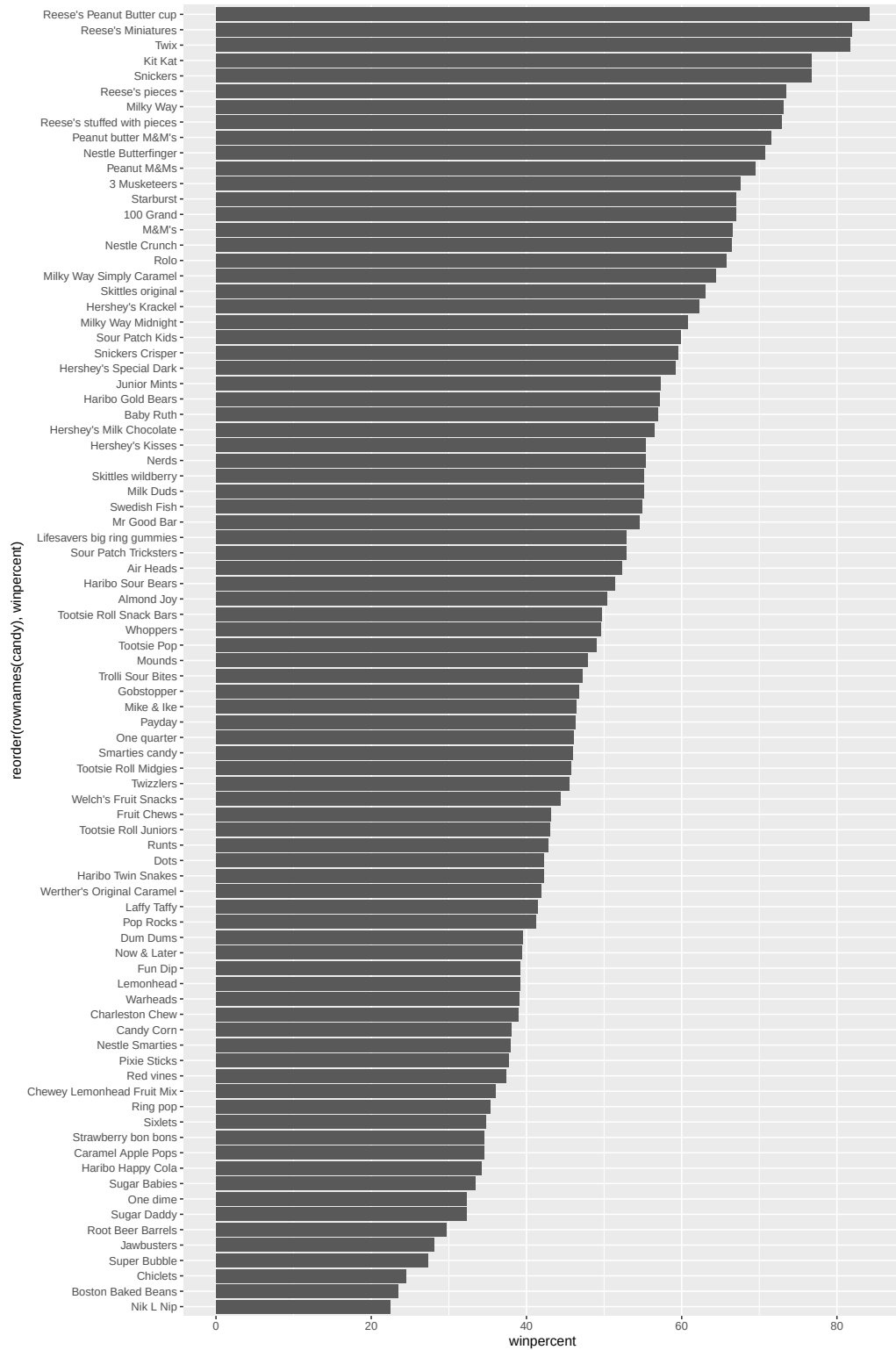
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +
  aes(winpercent, rownames(candy)) +geom_col()
```

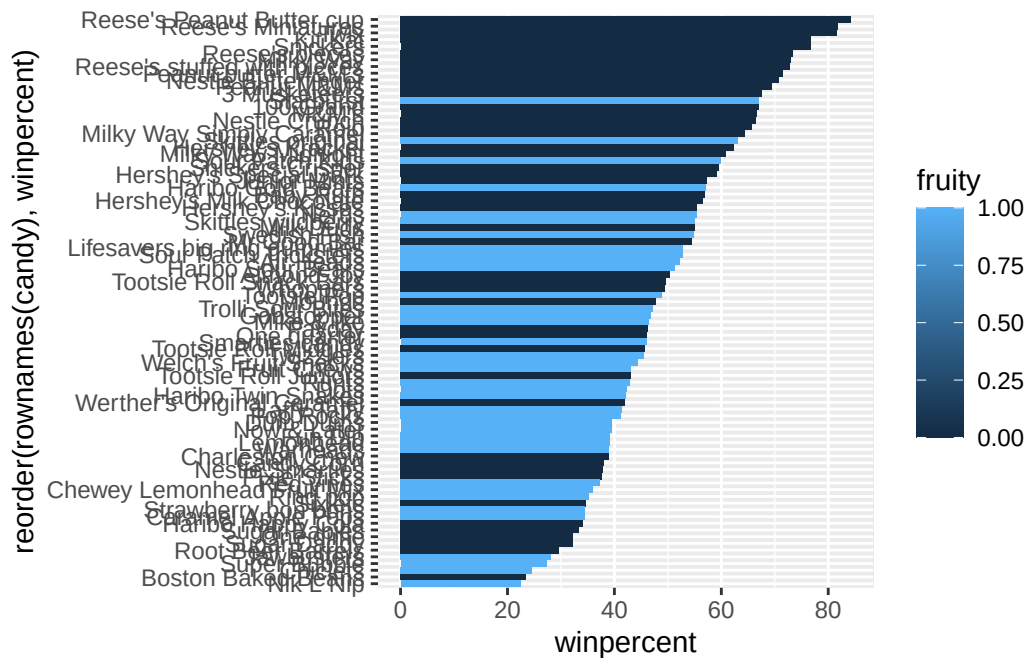


Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +geom_col()
```

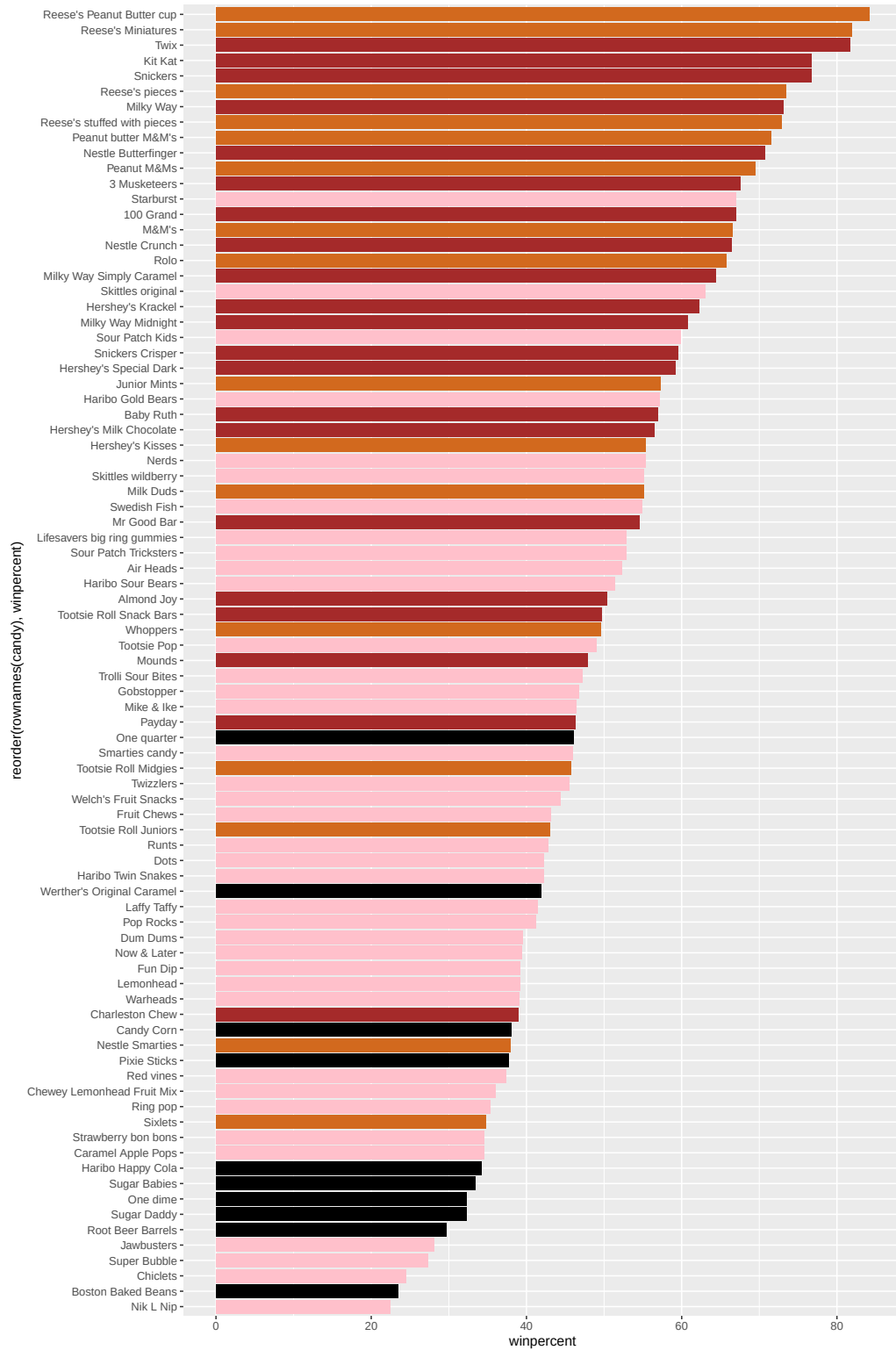


```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent), fill = fruity) + geom_col()
```



```
mycols <- rep("black", nrow(candy))

## Chocolate Candy
mycols[ as.logical (candy$chocolate) ] <- "chocolate"
#candy bars in brown
mycols [ as.logical (candy$bar)] <- "brown"
#fruity candy in pink
mycols [ as.logical(candy$fruity)] <- "pink"
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) + geom_col(fill = mycols)
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

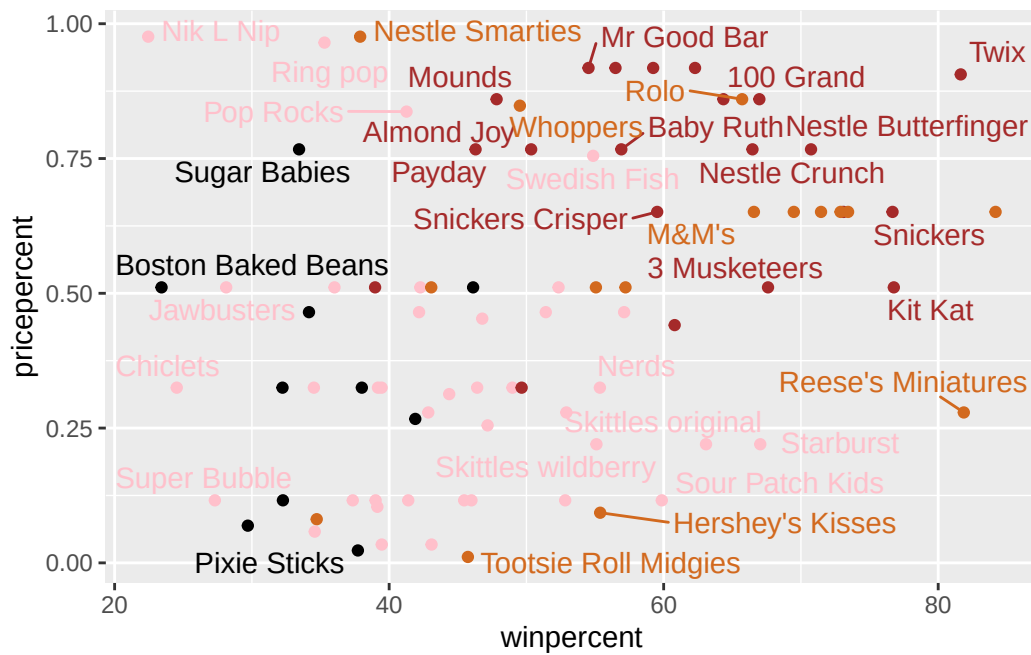
The best ranked fruity candy is Starburst.

Taking a look at pricepercent

Make a plot of winpercent vs. pricepercent

```
library(ggrepel)

ggplot(candy) +
  aes(winpercent,
       pricepercent,
       label = rownames(candy)) +
  geom_point(col = mycols) + geom_text_repel (col= mycols, max.overlaps = 10)
```



We can fix the label text overplotting with an add on package called **ggrepel** and it's `geom_text_repel()`.

```
mycols[as.logical(candy$fruity)] <- "red"
```

Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Peanut Butter Cup offers the most bang for your buck since it has the highest winpercent at 84.18.

```
candy[order(-candy$winpercent, candy$pricepercent),
       c("winpercent", "pricepercent")][1:10, ]
```

	winpercent	pricepercent
Reese's Peanut Butter cup	84.18029	0.651
Reese's Miniatures	81.86626	0.279
Twix	81.64291	0.906
Kit Kat	76.76860	0.511
Snickers	76.67378	0.651
Reese's pieces	73.43499	0.651
Milky Way	73.09956	0.651
Reese's stuffed with pieces	72.88790	0.651
Peanut butter M&M's	71.46505	0.651
Nestle Butterfinger	70.73564	0.767

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types from most to least popular is Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate.

```
head(candy[order(-candy$pricepercent),
                 c("pricepercent", "winpercent")], 5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

6 Exploring the correlation Structure

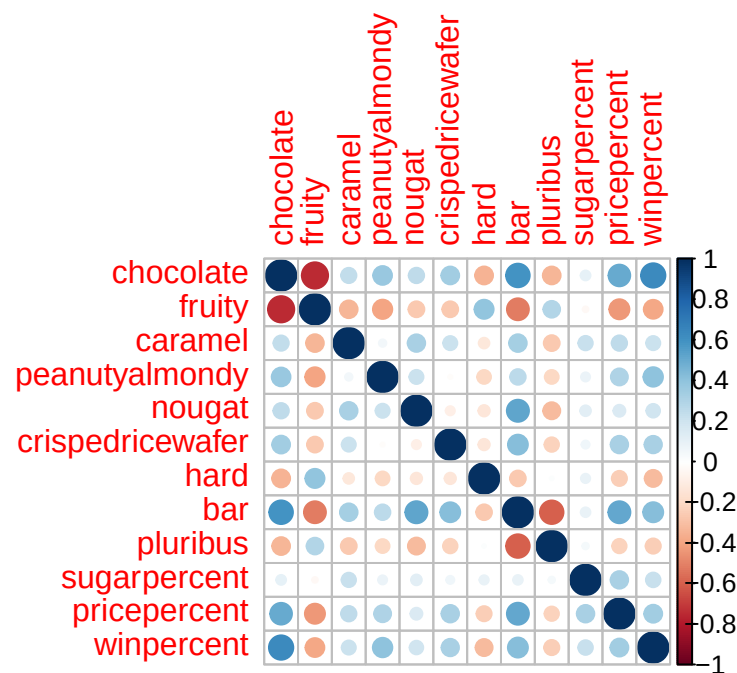
We can calculate the pair-wise correlation of all our columns

```
cij <- cor(candy)

library (corrplot)
```

corrplot 0.95 loaded

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

This shows us the variables chocolate and fruity are strongly anti correlated since it shows a big RED CIRCLE.

In this case we want to be sure to set `scale = True` argument for `prcomp()` because we have one variable `winpercent`, that is on a very different scale than all others and would otherwise dominate our PCA results.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)? HINT: Do you like chocolaty fruity candies?

The variables chocolate and fruity will most likely have the largest contributions to PC1 because they show a strong negative correlation and represent the main difference between candy types. This separation will drive the greatest variation in the data set.

Principal Component Analysis

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

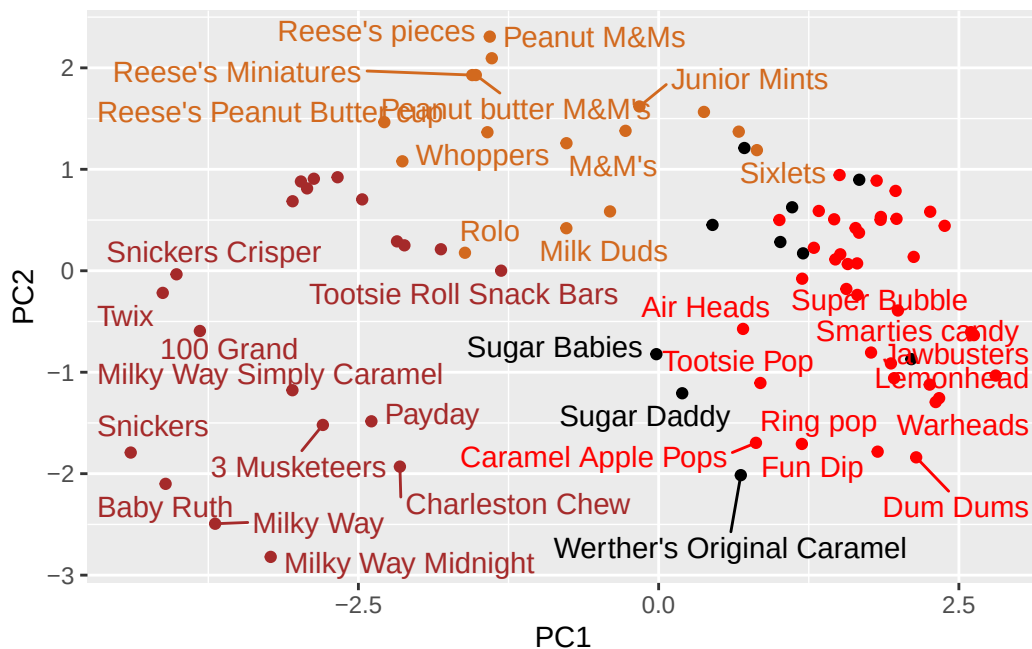
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First major result figure is the “score plot” of PC1 vs. PC2 - how the different candy are related to each other on our new PC axis:

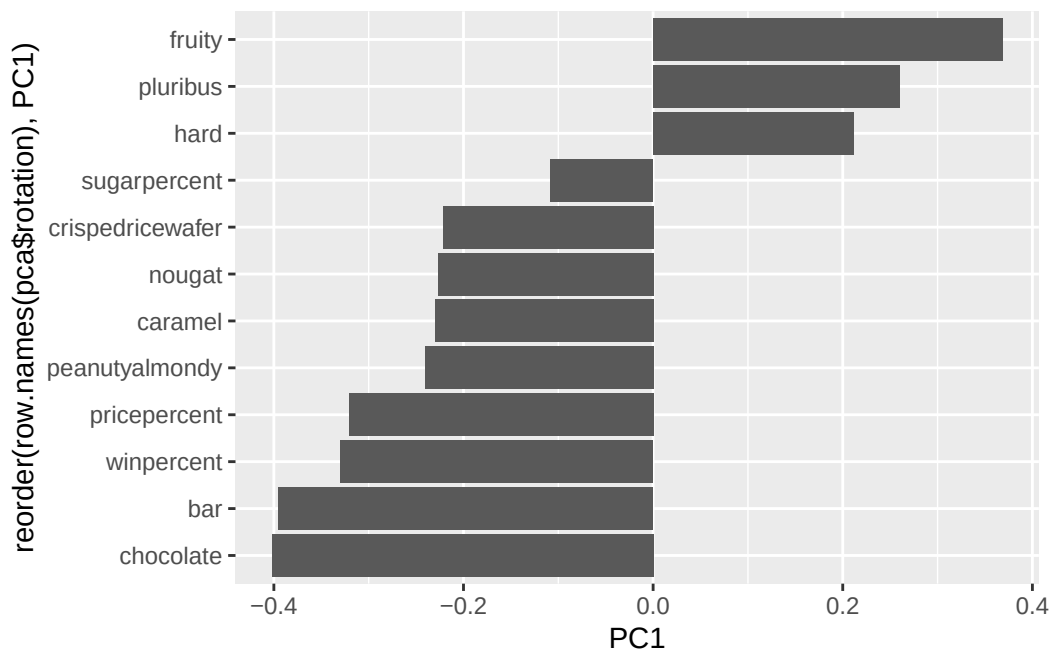
```
my_data <- cbind(candy, pca$x[,1:3])

ggplot(my_data) +
  aes(x = PC1, y = PC2,
      label = rownames(my_data)) +
  geom_point(col = mycols) +
  geom_text_repel(col = mycols)
```



The second major results figure from PCA.

```
ggplot(pca$rotation) + aes(PC1,
  reorder( row.names(pca$rotation), PC1)) + geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables that are picked up strongly by PC1 in the positive direction are fruity, pluribus, and hard. This means PC1 is partly separating candies that are fruity, come in many pieces as well as harder candies. This makes sense because earlier we saw that chocolate and fruity were strongly NEGATIVELY correlated. In the loading plot, fruity is positive and chocolate bars are negative, so PC1 is separating fruity candies from chocolate candies.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Winning candies tend to be chocolate based, has more sugar levels and often are in bar form. Looking at the correlation analysis, chocolate is positively associated with higher winpercent, while fruity candies tend to be less popular. The PCA results support this by showing that chocolate and fruity variable separate strongly along PC1, indicating they are key drivers of candy preference. Overall, visualization, correlation and PCA all point to chocolate candies being more favored rather than the fruity ones.